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ELEC 3200: System Modeling, Analysis and Control

Lecture 3: State-space Concept

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State-space concept

Notation Reminder

We will be looking at *dynamic systems* whose evolution *in time* is described by differential equations with external inputs.

We will not write the time variable t explicitly, so we use

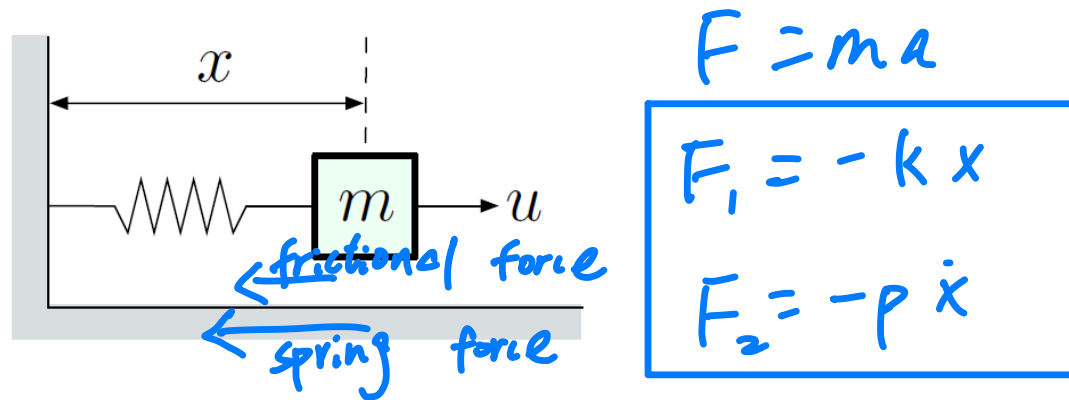
x instead of $x(t)$

$\dot{x}$ instead of $x'(t)$ or $\frac{dx}{dt}$

$\ddot{x}$ instead of $x''(t)$ or $\frac{d^2x}{dt^2}$

etc.

Example 1: Mass-Spring System



Newton's second law (translational motion):

$$\underbrace{F}_{\text{total force}} = ma = \text{spring force} + \text{friction} + \text{external force}$$
$$m\ddot{x} = -kx - p\dot{x} + u$$

$$\text{spring force} = -kx \quad (\text{Hooke's law})$$

$$\text{friction force} = -\rho\dot{x} \quad (\text{Stokes' law} — \text{linear drag, only an approximation!!})$$

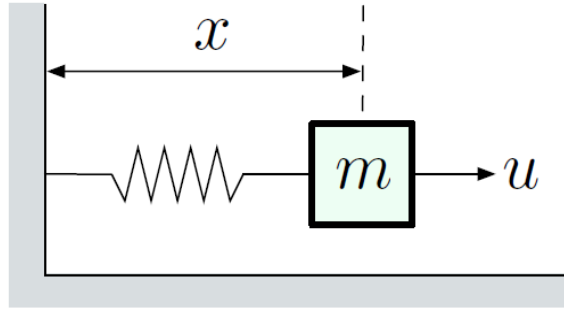
$$m\ddot{x} = -kx - \rho\dot{x} + u$$

Move $x, \dot{x}, \ddot{x}$ to the LHS, u to the RHS:

$$\ddot{x} + \frac{\rho}{m}\dot{x} + \frac{k}{m}x = \frac{u}{m} \quad \text{2nd-order linear ODE}$$

$$\begin{pmatrix} \dot{x} \\ \ddot{x} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{\rho}{m} \end{pmatrix} \begin{pmatrix} x \\ \dot{x} \end{pmatrix} + u \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix}$$

Example 1: Mass-Spring System



$$\ddot{x} + \frac{\rho}{m}\dot{x} + \frac{k}{m}x = \frac{u}{m} \quad \text{2nd-order linear ODE}$$

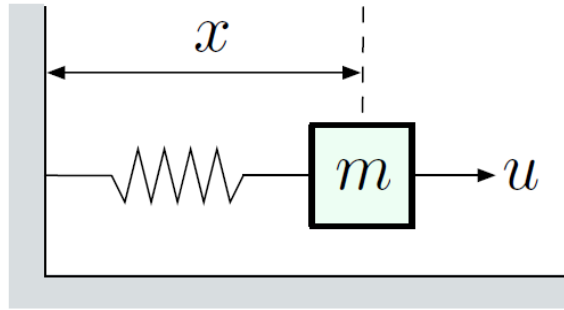
Canonical form: convert to a *system of 1st-order ODEs*

$$\dot{x} = v \quad (\text{definition of velocity})$$

$$\dot{v} = -\frac{\rho}{m}v - \frac{k}{m}x + \frac{1}{m}u$$

Example 1: Mass-Spring System

$$\dot{x} = Ax + Bu$$



State-space model: express in *matrix form*

system matrix!

equilibrium
 $x, \dot{x} = 0$ *change state*
 $= 0!$

$$\underbrace{\begin{pmatrix} \dot{x} \\ \dot{v} \end{pmatrix}}_{\dot{x}} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{\rho}{m} \end{pmatrix} \underbrace{\begin{pmatrix} x \\ v \end{pmatrix}}_x + \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix} u$$

vector!

Important: start reviewing your linear algebra *now!!*

- matrix-vector multiplication; eigenvalues and eigenvectors; etc.

$$\dot{x} = \begin{pmatrix} \end{pmatrix} \dot{x} + \begin{pmatrix} \end{pmatrix} u$$

General n -Dimensional State-Space Model

$$\text{state } x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n \quad \text{input } u = \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix} \in \mathbb{R}^m$$

$$\begin{pmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_n \end{pmatrix} = \begin{pmatrix} A \\ n \times n \\ \text{matrix} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} B \\ n \times m \\ \text{matrix} \end{pmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix}$$

$$\dot{x} = Ax + Bu$$

Partial Measurements

$$\text{state } x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n \quad \text{input } u = \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix} \in \mathbb{R}^m$$

$$\text{output } y = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix} \in \mathbb{R}^p \quad y = Cx \quad C - p \times n \text{ matrix}$$

control velocity

$$y = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}$$

get velocity

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

↑ output matrix

Example: if we only care about (or can only measure) x_1 , then

$$y = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}$$

$$y = x_1 = \begin{pmatrix} 1 & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

State-Space Models: Bottom Line

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

A – System Matrix

B – Input Matrix

C – Output matrix

State-space models are useful and convenient for writing down system models for different types of systems, in a unified manner.

All need to design !

Note: A more general case for the output equation is $y = Cx + Du$. Here, D is the feedthrough matrix, showing the effect of the input to the output. We will discuss this in L05 and L23.

State space concept

$$\begin{pmatrix} x \\ v \end{pmatrix}$$

Definition 2.1. The state variables of a system are a set of independent variables whose values at time t_0 , together with input for all $t \geq t_0$, determine the behavior of the system for all $t \geq t_0$.

- **State variables** are used to represent the states of a general system.
- The set of possible combinations of state variable values is called the **state space** of the system.
- The equations relating the current state of a system to its most recent input and past states are called the **state equations**: $\dot{x} = Ax + Bu$
- The equations expressing the values of the output variables in terms of the state variables and inputs are called the **output equations**: $y = Cx + Du$

Examples 1

$$\ddot{x} + a_1 \dot{x} + a_2 x = u$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$\underbrace{\begin{matrix} x & \dot{x} & \ddot{x} \end{matrix}}_{\text{state variables}}$

$$\ddot{x} = -a_1 \dot{x} - a_2 x + u$$

The state-space model?

$$\begin{pmatrix} \dot{x} \\ \ddot{x} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{pmatrix} \begin{pmatrix} x \\ \dot{x} \\ \ddot{x} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$

Examples 1

$$\dot{x} = Ax + Bu$$

$$\ddot{x} + a_1\dot{x} + a_2x = u$$

$$\bar{x} = \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} \Rightarrow \dot{\bar{x}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

A B

Examples 2

$$\begin{cases} \ddot{y} + a_1 \dot{y} + a_2 x = u_1 \\ \dot{x} + b_1 x + b_2 y = u_2 \end{cases}$$

$$\begin{aligned} \ddot{y} &= -a_1 \dot{y} - a_2 x + u_1 \\ \dot{x} &= -b_1 x - b_2 y + u_2 \end{aligned}$$

The state-space model?

$$\begin{pmatrix} \dot{y} \\ \ddot{y} \\ \dot{x} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -a_1 & -a_2 \\ b_2 & 0 & -b_1 \end{pmatrix} \begin{pmatrix} y \\ \dot{y} \\ x \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

Examples 2

$$\dot{x} = Ax + Bu$$

$$\begin{cases} \ddot{y} + a_1\dot{y} + a_2x = u_1 \\ \dot{x} + b_1x + b_2y = u_2 \end{cases}$$

$$\bar{x} = \begin{bmatrix} y \\ \dot{y} \\ x \end{bmatrix} \Rightarrow \dot{\bar{x}} = \begin{bmatrix} \dot{y} \\ \ddot{y} \\ \dot{x} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & -a_1 & -a_2 \\ -b_2 & 0 & -b_1 \end{bmatrix}}_A \begin{bmatrix} y \\ \dot{y} \\ x \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_B \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

How to set the state variables?

- Suppose that a differential equation model of a system is already obtained, the variables in the differential equations are the **input** $u(t)$ and the **internal variables** $v_1(t), \dots, v_p(t)$, and the **highest order of the derivatives** of $v_i(t)$ is q_i . Then we can choose:

$$v_i(t), \dot{v}_i(t), \ddot{v}_i(t), \dots, v_i^{(q_i-1)}(t) \text{ as the } \mathbf{state\ variables}. \quad i = 1, 2, \dots, p$$

In this case, the total number of state variables is $\sum_{i=1}^p q_i$

After the **state variables** are chosen, usually named $x_1(t), x_2(t), \dots, x_n(t)$, we put them into a vector $\mathbf{x}(t)$. This vector is called a **state vector**.

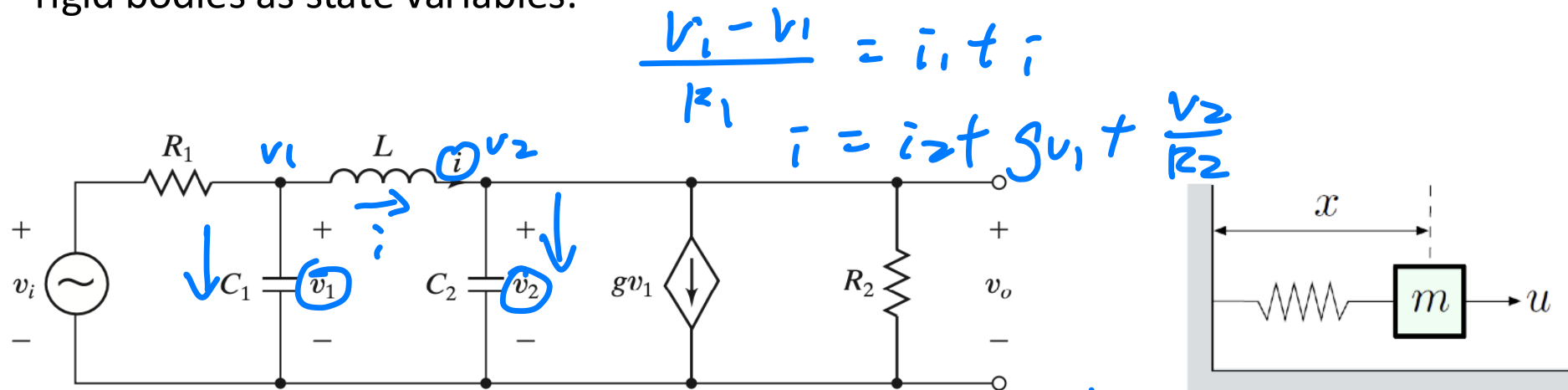
$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \in \mathbb{R}^n$$

For example, in our previous exercise

$$\begin{cases} \ddot{y} + a_1 \dot{y} + a_2 x = u_1 \\ \dot{x} + b_1 x + b_2 y = u_2 \end{cases} \quad \text{We set } \bar{\mathbf{x}} = \begin{bmatrix} y \\ \dot{y} \\ x \end{bmatrix} \text{ as the state vector.}$$

How to set the state variables?

- In many situations, the state variables can be chosen intuitively:
 - For **electrical circuits**, we can always choose the **voltages across independent capacitors** and the **currents through independent inductors** as state variables.
 - For **mechanical systems**, we can always choose the **positions and velocities** of independent rigid bodies as state variables.



$$i_1 = C_1 \frac{dv_1}{dt}$$

$$i_2 = C_2 \frac{dv_2}{dt}$$

$$v_1 - v_2 = L \frac{di}{dt}$$

$$\begin{pmatrix} \dot{v}_1 \\ \dot{v}_2 \end{pmatrix} = \begin{pmatrix} \cdot & \cdot \\ \pm & \mp \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} v_i$$

Examples will be given in the next lecture $v_o = \begin{pmatrix} \cdot & \cdot \end{pmatrix} v_i$

Review & Exercise

- What is “state variables”?
- What is “state space”? $\dot{x} = Ax + Bu$ $\bar{y} = Cx$
- Can you write out the “state equation” and “output equation”?
- Can you establish the state-space model of a mass-spring system?
- Can you change the differential equation to state-space model?
- Can you list some skills to set the state variables? e.g., in the mechanical system, and in the electrical system.
- Exercise: Establish the state-space-model for the free-falling object: (i) no friction; (ii) the friction force is proportional to the velocity.

(i) No friction

Hint:

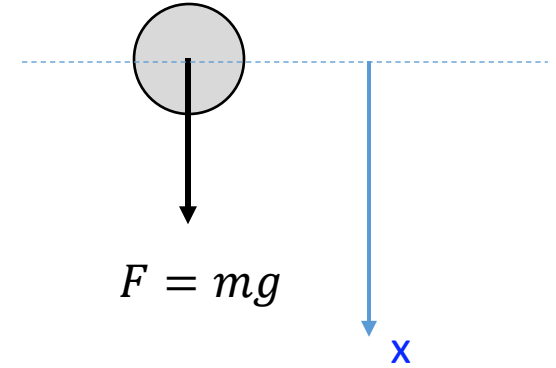
$$\begin{aligned} F &= ma = m\ddot{x} \\ v &= \dot{x} \end{aligned} \quad \Rightarrow \quad \begin{aligned} \ddot{x} &= g \\ \dot{x} &= v \end{aligned}$$

State variable: $\bar{x} = \begin{bmatrix} x \\ v \end{bmatrix}$ or $\bar{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$

State-space model: $\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F$

$\begin{bmatrix} 0 \\ g \end{bmatrix}$

or $\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F$



(ii) Friction force is proportional to the velocity

Hint:

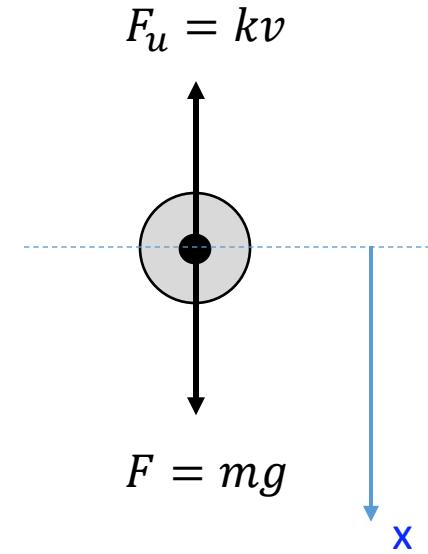
$$\begin{aligned} F - F_u &= ma = m\ddot{x} \\ v &= \dot{x} \end{aligned} \quad \Rightarrow \quad \begin{aligned} \ddot{x} &= g - \frac{k}{m}\dot{x} \\ \dot{x} &= v \end{aligned}$$

State variable: $\bar{x} = \begin{bmatrix} x \\ v \end{bmatrix}$ or $\bar{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$

State-space model: $\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{k}{m} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F$

$\begin{bmatrix} 0 \\ g \end{bmatrix}$

or $\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{k}{m} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F$



Appendix

Matrices

An $m \times n$ matrix: the m rows are horizontal and the n columns are vertical. Each element of a matrix is often denoted by a variable with two subscripts. For example, $a_{2,1}$ represents the element at the second row and first column of the matrix.

$$\begin{matrix} & \begin{matrix} 1 & 2 & \dots & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{matrix} & \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{matrix}$$

Overview of a matrix size

Name	Size	Example	Description
Row vector	$1 \times n$	$\begin{bmatrix} 3 & 7 & 2 \end{bmatrix}$	A matrix with one row, sometimes used to represent a vector
Column vector	$n \times 1$	$\begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$	A matrix with one column, sometimes used to represent a vector
Square matrix	$n \times n$	$\begin{bmatrix} 9 & 13 & 5 \\ 1 & 11 & 7 \\ 2 & 6 & 3 \end{bmatrix}$	A matrix with the same number of rows and columns, sometimes used to represent a linear transformation from a vector space to itself, such as reflection, rotation, or shearing.

Addition

The *sum* $\mathbf{A} + \mathbf{B}$ of two m -by- n matrices $\mathbf{A}$ and $\mathbf{B}$ is calculated entrywise:

$$(\mathbf{A} + \mathbf{B})_{i,j} = \mathbf{A}_{i,j} + \mathbf{B}_{i,j}, \text{ where } 1 \leq i \leq m \text{ and } 1 \leq j \leq n.$$

For example,

$$\begin{bmatrix} 1 & 3 & 1 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 5 \\ 7 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 1+0 & 3+0 & 1+5 \\ 1+7 & 0+5 & 0+0 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 6 \\ 8 & 5 & 0 \end{bmatrix}$$

Scalar multiplication

The product $c\mathbf{A}$ of a number c (also called a [scalar](#) in this context) and a matrix $\mathbf{A}$ is computed by multiplying every entry of $\mathbf{A}$ by c :

$$(c\mathbf{A})_{i,j} = c \cdot \mathbf{A}_{i,j}.$$

This operation is called *scalar multiplication*, but its result is not named "scalar product" to avoid confusion, since "scalar product" is often used as a synonym for "[inner product](#)". For example:

$$2 \cdot \begin{bmatrix} 1 & 8 & -3 \\ 4 & -2 & 5 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 & 2 \cdot 8 & 2 \cdot -3 \\ 2 \cdot 4 & 2 \cdot -2 & 2 \cdot 5 \end{bmatrix} = \begin{bmatrix} 2 & 16 & -6 \\ 8 & -4 & 10 \end{bmatrix}$$

Subtraction

The subtraction of two $m \times n$ matrices is defined by composing matrix addition with scalar multiplication by -1 :

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-1) \cdot \mathbf{B}$$

Transposition

The *transpose* of an m -by- n matrix $\mathbf{A}$ is the n -by- m matrix $\mathbf{A}^T$ (also denoted $\mathbf{A}^{\text{tr}}$ or ${}^t\mathbf{A}$) formed by turning rows into columns and vice versa:

$$(\mathbf{A}^T)_{i,j} = \mathbf{A}_{j,i}.$$

For example:

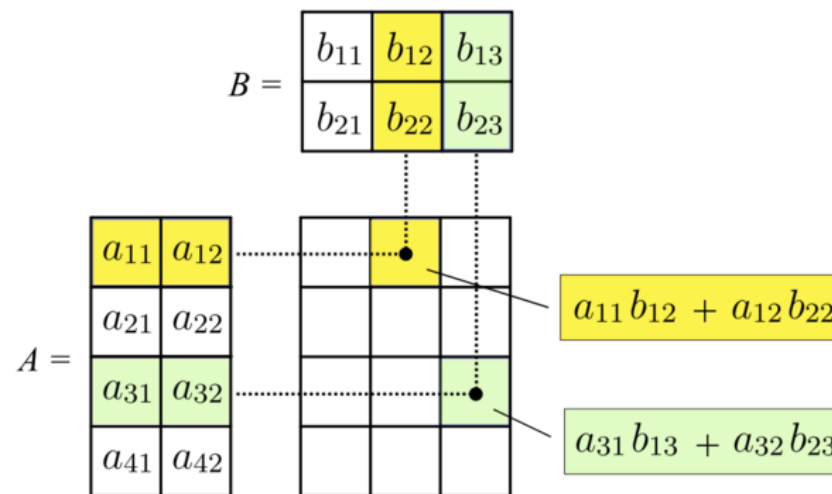
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -6 & 7 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 2 & -6 \\ 3 & 7 \end{bmatrix}$$

Familiar properties of numbers extend to these operations on matrices: for example, addition is [commutative](#), that is, the matrix sum does not depend on the order of the summands: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.^[9]

The transpose is compatible with addition and scalar multiplication, as expressed by $(c\mathbf{A})^T = c(\mathbf{A}^T)$ and $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$. Finally, $(\mathbf{A}^T)^T = \mathbf{A}$.

Matrix multiplication

$$[\mathbf{AB}]_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j} = \sum_{r=1}^n a_{i,r}b_{r,j},$$



Matrix multiplication satisfies the rules $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ ([associativity](#)), and $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$ as well as $\mathbf{C}(\mathbf{A} + \mathbf{B}) = \mathbf{CA} + \mathbf{CB}$ (left and right [distributivity](#)), whenever the size of the matrices is such that the various products are defined.^[12] The product $\mathbf{AB}$ may be defined without $\mathbf{BA}$ being defined, namely if $\mathbf{A}$ and $\mathbf{B}$ are m -by- n and n -by- k matrices, respectively, and $m \neq k$. Even if both products are defined, they generally need not be equal, that is:

$$\mathbf{AB} \neq \mathbf{BA},$$

In other words, matrix multiplication is not [commutative](#), in marked contrast to (rational, real, or complex) numbers, whose product is independent of the order of the factors.^[10] An example of two matrices not commuting with each other is:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 3 \end{bmatrix},$$

whereas

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix}.$$

Determinant

In **mathematics**, the **determinant** is a **scalar value** that is a **function** of the entries of a **square matrix**. The determinant of a matrix A is commonly denoted $\det(A)$, $\det A$, or $|A|$. Its value characterizes some properties of the matrix and the **linear map** represented by the matrix. In particular, the determinant is nonzero **if and only if** the matrix is **invertible** and the linear map represented by the matrix is an **isomorphism**. The determinant of a product of matrices is the product of their determinants (which follows directly from the above properties).

The determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

and the determinant of a 3×3 matrix is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Eigenvalues and eigenvectors [[edit](#)]

Main article: [Eigenvalues and eigenvectors](#)

A number λ and a non-zero vector $\mathbf{v}$ satisfying

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

are called an *eigenvalue* and an *eigenvector* of $\mathbf{A}$, respectively.^{[39][40]} The number λ is an eigenvalue of an $n \times n$ -matrix $\mathbf{A}$ if and only if $\mathbf{A} - \lambda\mathbf{I}_n$ is not invertible, which is [equivalent](#) to

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0.^[41]$$